KetoTap

Macro Calculator — Feature Requirements

Version 1.0 | March 2026 | For Claude Code

Field	Value
Feature Name	Keto Macro Target Calculator
Feature ID	KT-FEAT-003
Author	KetoTap Product
Status	Ready for Development
Target Screen	New screen: /calculator (accessible from Dashboard)
Design System	DESIGN_SYSTEM.md — avocado light theme
Primary User	New and existing keto users who want personalised macro targets

1. Overview & Purpose

This document specifies all requirements for the Keto Macro Target Calculator, a new feature in KetoTap that allows users to determine their optimal daily fat, protein, and carbohydrate targets based on their body stats, activity level, and personal goals.

The calculator removes the guesswork from starting or adjusting a ketogenic diet. Rather than asking users to search for generic advice online, KetoTap will compute scientifically-grounded personalised targets and apply them directly to the user's macro tracking dashboard.

This feature must feel like a natural extension of the existing KetoTap UI. It should follow the DESIGN_SYSTEM.md specification exactly — avocado light theme, Nunito font, green/brown/accent palette, and the standard card/chip component patterns.

2. User Stories

The following user stories define the intended experience from the user's perspective.

- As a new user, I want to enter my height, weight, age, and activity level so that KetoTap can calculate a personalised calorie and macro target for me.

- As a user who wants to lose fat, I want to select a fat-loss goal and see a calorie deficit applied automatically to my targets.
- As a user doing a lean bulk, I want to select a muscle-gain goal and see a calorie surplus applied to my targets.
- As a power user, I want to manually override the fat/protein/carb percentage split so that I can follow a custom ratio that my doctor or nutritionist has recommended.
- As a user with a specific calorie goal in mind, I want to enter a custom deficit or surplus in calories rather than choosing a preset goal.
- As any user, I want to see a clear summary of my calculated targets — calories, fat grams, protein grams, and carb grams — before I confirm and apply them to my dashboard.
- As a returning user, I want my previously saved stats to pre-fill the calculator so I don't have to re-enter everything.

3. Calculation Logic & Formulas

This section defines the exact math the app must implement. All formulas must be encapsulated in a standalone utility file (e.g. macroCalculator.ts or macroCalculator.js) that is independently testable. No calculation logic should live inside UI components.

3.1 Step 1 — Basal Metabolic Rate (BMR)

BMR is the number of calories the body burns at complete rest. Use the Mifflin-St Jeor equation, which is the most accurate formula for general populations and is preferred over the older Harris-Benedict equation.

Mifflin-St Jeor Equation

```
// For males:

BMR = (10 x weightKg) + (6.25 x heightCm) - (5 x ageYears) + 5



// For females:

BMR = (10 x weightKg) + (6.25 x heightCm) - (5 x ageYears) - 161
```

Unit conversion: if the user enters imperial units (lbs / inches), convert before applying the formula.

```
weightKg = weightLbs / 2.2046

heightCm = heightInches * 2.54
```

3.2 Step 2 — Total Daily Energy Expenditure (TDEE)

TDEE is the user's real-world daily calorie burn, accounting for physical activity. Multiply BMR by the appropriate activity multiplier from the table below.

Activity Level	Multiplier	Description
Sedentary	1.2	Desk job, little or no exercise
Lightly Active	1.375	Light exercise 1-3 days/week
Moderately Active	1.55	Moderate exercise 3-5 days/week
Very Active	1.725	Hard exercise 6-7 days/week
Extremely Active	1.9	Physical job + hard daily exercise

```
TDEE = BMR x activityMultiplier
```

3.3 Step 3 — Goal-Based Calorie Adjustment

Apply a calorie adjustment to TDEE based on the user's selected goal. This produces the Target Daily Calories value that all macro calculations are based on.

Goal	Adjustment	Notes
Lose Fat	-500 kcal/day	Safe deficit for ~0.5kg/week fat loss. Never go below 1200 kcal (female) or 1500 kcal (male).
Gain Muscle (Lean Bulk)	+250 kcal/day	Conservative surplus to minimise fat gain while in ketosis.
Custom Deficit / Surplus	User-entered value	Accept a value from -1000 to +500 kcal/day. Show a warning if the deficit exceeds -750 kcal/day.

```
targetCalories = TDEE + goalAdjustment

// Safety floor:
if (sex === 'female') targetCalories = Math.max(targetCalories, 1200)
if (sex === 'male')   targetCalories = Math.max(targetCalories, 1500)
```

Always display the safety floor warning if the floor is triggered. Example: "Your calculated target was 1,050 kcal. We have raised it to 1,200 kcal to maintain safe nutritional intake."

3.4 Step 4 — Macro Split

Convert targetCalories into gram targets for each macro using the user's chosen ratio. The default preset is Standard Keto. Users may override any percentage via sliders (see UI section).

Preset	Fat %	Protein % | Carbs %
Standard Keto	70%	25% | 5%
Strict / Therapeutic	75%	20% | 5%
Targeted Keto (TKT)	65%	25% | 10%
Custom (Manual Override)	User-set	User-set | User-set

Caloric density constants (fixed values — do not make these configurable):

```
FAT_KCAL_PER_GRAM     = 9

PROTEIN_KCAL_PER_GRAM = 4

CARB_KCAL_PER_GRAM    = 4
```

Gram conversion formulas:

```
fatGrams     = (targetCalories x fatPct)     / FAT_KCAL_PER_GRAM

proteinGrams = (targetCalories x proteinPct) / PROTEIN_KCAL_PER_GRAM

carbGrams    = (targetCalories x carbPct)    / CARB_KCAL_PER_GRAM



// All gram values should be rounded to the nearest whole number.

// fatPct, proteinPct, carbPct are decimals (e.g. 0.70, 0.25, 0.05).

// They must always sum to exactly 1.0 — enforce this in the UI.
```

3.5 Validation Rules

The following validation rules must be enforced before any calculation is performed. Display inline error messages in the form — never use alert() or toast for validation errors.

Field	Validation Rule
Weight	Required. Must be a positive number. Min: 30 kg (66 lbs). Max: 300 kg (661 lbs).
Height	Required. Must be a positive number. Min: 100 cm (39 in). Max: 250 cm (98 in).
Age	Required. Integer only. Min: 16. Max: 100.
Biological Sex	Required. Must be 'male' or 'female'. (Mifflin-St Jeor requires binary input.)
Activity Level	Required. Must be one of the 5 defined levels.
Goal	Required. Must be one of: lose_fat, gain_muscle, custom.
Custom Adjustment	Required if goal = custom. Integer. Range: -1000 to +500 kcal.
Macro percentages	Must sum to exactly 100%. Enforce via slider lock (see UI section 5.3).

4. Data Model

Define the following TypeScript interfaces (or equivalent JS objects). Store user stats in the existing user profile object; store computed results separately so they can be recalculated without losing the inputs.

```
interface UserStats {
  weightKg:        number;
  heightCm:        number;
  ageYears:        number;
  biologicalSex:   'male' | 'female';
  activityLevel:   'sedentary' | 'light' | 'moderate' | 'very' | 'extreme';
  units:           'metric' | 'imperial';
}
```

```
interface GoalConfig {
  goal:             'lose_fat' | 'gain_muscle' | 'custom';
  customAdjustment: number | null;  // kcal, only used when goal = 'custom'
  macroPreset:      'standard' | 'strict' | 'targeted' | 'custom';
```

```
  fatPct:           number;         // 0.0 - 1.0
  proteinPct:       number;         // 0.0 - 1.0
  carbPct:          number;         // 0.0 - 1.0
}

interface MacroTargets {
  bmr:             number;   // kcal
  tdee:            number;   // kcal
  targetCalories:  number;   // kcal (post-adjustment, post-safety-floor)
  fatGrams:        number;
  proteinGrams:    number;
  carbGrams:       number;
  safetyFloorHit:  boolean;
  calculatedAt:    string;   // ISO timestamp
}
```

5. UI / UX Requirements

The calculator is a multi-step flow presented as a single scrollable screen on mobile, or as a stepped card wizard on tablet/desktop. It must follow DESIGN_SYSTEM.md exactly.

5.1 Entry Point

- Add a "Calculate My Targets" button/card to the dashboard, below the macro chips section.
- Style it as a full-width banner card using greenPale background and a green border, with an avocado emoji and the text "Calculate your personalised keto targets".
- Also accessible from Settings > Nutrition Goals.

5.2 Step 1 — Body Stats Form

- Fields: Weight, Height, Age, Biological Sex (toggle), Activity Level (segmented or dropdown).

- Provide a metric / imperial unit toggle at the top of the form. Switching units should convert and re-display current values without clearing them.
- Use the standard KetoTap InputField component (see DESIGN_SYSTEM.md) for all text inputs.
- Activity Level should render as a vertically stacked radio card list, each showing the level name and a one-line description. Active selection uses greenPale background and a green left border.
- Pre-fill all fields from the saved UserStats object if it exists.

5.3 Step 2 — Goal & Macro Ratio

- Goal selection: three large tap cards (Lose Fat, Gain Muscle, Custom). Each card shows an emoji, title, and one-line description. Active card uses a coloured top accent line matching the macro colour scheme (brown for lose fat, green for gain muscle, accent gold for custom).
- Custom goal: reveal a numeric stepper input for the calorie adjustment. Show the TDEE value prominently so users understand the baseline. Show a warning banner if the entered deficit exceeds -750 kcal.
- Macro Preset selector: show 3 preset chips (Standard, Strict, Targeted). Selecting a preset locks the sliders to that ratio. Selecting "Custom" unlocks all three sliders.
- Macro sliders: three sliders for fat%, protein%, carb%. Use the macro colour for each slider track (brown, green, gold). Implement a locked-sum constraint: when the user drags one slider, the other two adjust proportionally so the total always equals 100%.
- Show live gram previews next to each slider as the user adjusts them, so they can see "142g fat" etc. in real time.

5.4 Step 3 — Results Screen

The results screen is the most important screen in the flow. It must be visually rewarding and clearly legible.

- Show a large Keto Score-style hero number for Target Daily Calories (e.g. "1,840 kcal/day") at the top of the results card.
- Below it, show the four macro chips (fat, protein, carbs, calories) using the exact MacroChip component from the dashboard, with the calculated gram values as both current and goal (so the ring shows 100% filled as a target preview).
- Show a calculation breakdown section (collapsible) that displays BMR, TDEE, goal adjustment, and final calories in a step-by-step list so users can understand how their number was arrived at.
- Show a prominent "Apply to Dashboard" CTA button (primary green gradient button). Tapping this saves the MacroTargets object to the user's profile and updates the dashboard macro goals.
- Show a secondary "Recalculate" link that returns to Step 1 without clearing the form.
- If safetyFloorHit is true, display the safety floor warning banner (amber/accent colour) above the results.

5.5 Accessibility & Edge Cases

- All form inputs must have visible labels (not just placeholder text).
- Error messages must appear inline, directly below the relevant field, in red text.
- The "Apply to Dashboard" action must show a confirmation state (e.g. a checkmark animation) before navigating away.
- If a user navigates away mid-flow without applying, prompt with a confirmation dialog: "Leave without saving your targets?"
- On re-entry to the calculator, show the most recently calculated results alongside the option to recalculate.

6. File & Architecture Guide

Claude Code should create or modify the following files. This structure assumes a React Native or React web app.

File	Purpose
src/utils/macroCalculator.ts	Pure calculation logic. All formulas from Section 3. No imports from React or UI. Must be 100% unit-testable.
src/utils/macroCalculator.test.ts	Unit tests covering all formulas, edge cases, and the safety floor logic.
src/screens/CalculatorScreen.tsx	Multi-step screen wrapper. Manages step state (stats -> goal -> results).
src/components/calculator/StatsForm.tsx	Step 1 form component — body stats inputs.
src/components/calculator/GoalForm.tsx	Step 2 form component — goal selection and macro ratio sliders.
src/components/calculator/ResultsCard.tsx	Step 3 results display with macro chips and Apply CTA.
src/components/calculator/MacroSlider.tsx	Reusable slider with locked-sum constraint logic.
src/types/calculator.ts	TypeScript interfaces: UserStats, GoalConfig, MacroTargets.
src/store/userProfileStore.ts	Add fields for UserStats, GoalConfig, MacroTargets. Persist to AsyncStorage or equivalent.

7. Test Cases

The following test cases must pass before this feature is considered complete. Claude Code should implement these as unit tests in macroCalculator.test.ts.

Test ID	Input	Expected Output
TC-001	Male, 80kg, 180cm, 30y, Sedentary, Lose Fat, Standard Keto	BMR=1814, TDEE=2177, Target=1677, Fat=131g, Protein=105g, Carbs=21g
TC-002	Female, 65kg, 165cm, 28y, Moderate, Maintain, Standard Keto	BMR=1434, TDEE=2222, Target=2222, Fat=173g, Protein=139g, Carbs=28g
TC-003	Male, 75kg, 175cm, 35y, Very Active, Gain Muscle, Strict Keto	BMR=1731, TDEE=2986, Target=3236, Fat=269g, Protein=162g, Carbs=40g
TC-004	Female, 55kg, 160cm, 45y, Light, Custom -800 kcal, Custom 65/30/5	Safety floor triggered (1200 kcal). Warning displayed. Fat=87g, Protein=90g, Carbs=15g
TC-005	Macro sliders: fat=70, protein=25, carbs=5 — user drags fat to 80	Protein and carbs reduce proportionally. Total remains 100%.
TC-006	Imperial input: 175 lbs, 5ft 10in — converted to metric	weightKg=79.4, heightCm=177.8. Calculation proceeds correctly.

8. Out of Scope

The following are explicitly excluded from this feature version to keep scope focused. They may be addressed in future iterations.

- Body fat percentage input and lean mass-based protein calculation (planned for v1.1).
- Integration with wearables or fitness trackers for automatic TDEE updates.
- Macro cycling or carb cycling schedules.
- Calorie targets for specific days of the week (e.g. workout vs. rest days).
- AI-generated meal plan suggestions based on calculated targets.
- Medical or clinical keto protocols (epilepsy, cancer therapy) — these require physician oversight and are out of scope for a consumer app.

9. Acceptance Criteria

This feature is considered complete when all of the following are true:

- All 6 test cases in Section 7 pass.
- The calculator entry point is visible on the dashboard.
- All three goal types (lose fat, gain muscle, custom) produce correct calorie targets.
- The safety calorie floor triggers and displays its warning for both male and female thresholds.
- Manual macro override sliders enforce the 100% sum constraint at all times.
- Applying results from the calculator updates the dashboard macro chip goals in real time.
- All UI components match DESIGN_SYSTEM.md (colours, fonts, border radii, spacing).
- The feature has been tested on both iOS and Android (or desktop and mobile web, depending on platform).

Questions about this spec? Refer to DESIGN_SYSTEM.md or the KetoTap product brief. **ketotap.app**